The prodigal nominative: an Andic story.

Jérémy Pasquereau, UMass (Amherst)*

May 15, 2015

1 Introduction

- Andic languages: Andi, **Akhvakh**, **Bagvalal**, Botlikh, **Chamalal**, **Godoberi**, **Karata**, and **Tindi**
- data from (Creissels 2009), (Pasquereau 2010), (Pasquereau 2011), and from available dictionaries: (Magomedova 1999), (Magomedova 2003), (Magomedova 2004), (Magomedova and Abdulaeva 2007), (Magomedova and Khalidova 2001), (Saidova 2006)

Figure 1: Andic languages in Daghestan

(1) Claim
The grammars of Andic languages have a constraint against nominative-less case frames (*-NOM)

- case frames with nominative case are expected (`regular')
- case frames <u>without</u> a nominative case are unexpected (`exceptional')
  - <ERG>

(2) imo-l biɬ'e χ̄ã-da idja Karata
father$_0$-ERG too_much snore-IPF COP
Father is snoring a lot.

  - <ERG, GEN>

(3) hu-s̄^{w}-e daru-ɬ̄i-īi ħalbix̄il-āri Akhvakh
DEM-M$_0$-ERG medicine-N$_0$-GEN try-PF
He tried the medicine.

  - <ERG, LOC, LOC>

(4) ho-šu-l di-č'o baʕilo-īi pũš-ā Karata
DEM-M$_0$-ERG 1SG$_0$-TPL[LOC] face$_0$-TPL[LOC] pulverise-CAUS.PF
He punched me in the face very hard.

*Part of this work was done in collaboration with Denis Creissels, *Université Lumière Lyon II*.

Q1: If there is a constraint against nominative-less case frames, how can they arise?

Q2: Once they have arisen, what is the effect of the constraint?

Q3: How does that constraint interact with the semantics of case?

2 Background of Andic morphosyntax

- the nominative (a.k.a. absolutive) case is the unmarked case, it is also used as the citation form of nouns
- verbs agree in gender and number (a.k.a. class) with the nominative argument only
- the other cases are marked by suffixes (they attach directly to the unmarked stem or to an `augmented' stem called *oblique stem*)
- spatial case endings consist of 2 formatives: a topological marker (TPL) and a directionality marker (DIR) (as with English *onto*: *on* `TPL' + *to* `DIR')
- ergative case alignment

(5) a. b-oʁoše ād-o atobusī-ɬi-g-e b-eq'-īri Akhvakh
H$^+$-most person-PL bus-N$_0$-TPL-LOC H$^+$-come-PF.H$^+$

Most people came by bus.

b. wašo-de istaka b-iq'^{w}-aj-ē godi
boy$_0$-ERG glass N-break-CAUS-ADV.N COP.N

The boy broke the glass.

- constituent order plays no role in the expression of argument structure: valency patterns are defined in terms of case frames

3 Origin of <ERG, LOC> and <ERG, ALL>

- these case frames are found with the verbs of damage to the body *bite*, *sting*, *pinch*, and the `peer verbs' *look*, *listen*.
- <ERG, LOC>

(6) di-g-e ī'iž^{w}ali-de č'in-āri Akhvakh
1SG$_0$-TPL-LOC wasp$_0$-ERG sting-PF

A wasp stung me.

(7) o-š^w-i č'un-o di-č'-i Tindi
DEM-M$_0$-ERG pinch-PF 1SG$_0$-TPL-LOC/ALL

He pinched me.

- <ERG, ALL>

(8) wašo-de di-g-a eq̄-ari Akhvakh
boy$_0$-ERG 1SG$_0$-TPL-ALL look-PF

The boy looked at me.

(9) Hypothesis:
<ERG, ~~NOM~~, LOC/ALL> → <ERG, LOC/ALL>

- the NOM argument can be lost in a number of ways, I focus on incorporation with the V: `X_{ERG} [applies/holds Y_{NOM}] on $Z_{LOC/ALL}$'
- incorporation in Tindi:
  - Y_{NOM} = *eq̄^{w}a* `stinger' in *eq̄$^w$a k̄$^w$ēɬ$^j$a* `sting', lit. `hit the stinger'
  - Y_{NOM} = *saldi* `teeth' in *saldi bix̄$^j$iɬ$^j$a* `bite', lit. `hold the teeth'
- in the next 2 sections, we will see
  - evidence of incorporation, and
  - evidence of the constraint against nominativeless case frames in the re-emergence of a nominative-marked argument

4 *listen* across Andic languages

- *listen* is used in 3 case frames across Andic languages
  - <ERG, NOM, ALL> in Godoberi *hãt'uk'ja riki*, Y_{NOM} = *hãt'uk'ja* `ear'

(10) wašu-di imu-q̄-i hãt'uk^{j}a r-ik̄-i r-uk̄-ida Godoberi
son$_0$-ERG father$_0$-TPL-LOC/ALL ear N^+-hold-INF N^+-must-IPF
The son must listen to his father.

  - <ERG, ALL> in Bagvalal *aštila*, Chamalal *wołuk'la*, Tindi *anix̄^{j}iɬ^{j}a*

(11) di-q̄-a anix̄^{j}-ā hik^{j}i o-š̌^{w}-i Tindi
1SG$_0$-TPL-LOC/ALL listen-IPF NEG DEM-M$_0$-ERG
He does not listen to me.

  - <NOM, ALL> in Akhvakh *hãdax̄uruɪa*, Karata *ãdukała*

(12) waša imo-g-a hãdax̄-ari Akhvakh
boy father$_0$-TPL-ALL listen-PF
The boy listened to his father.

(13) waša imo-χa-r ãduk-e Karata
boy father$_0$-TPL-ALL listen-PF
The boy listened to his father.

(14) Etymology of `listen' in Akhvakh
hãde b-ix̄uruɪa ≻ hãdax̄uruɪa
ear hold listen

- reconstruction of the evolution of the case frame associated to *listen*
  - Initial stage: <ERG, NOM, ALL> Godoberi
  - Nomativeless stage: <ERG, ALL> Bagvalal, Chamalal, Tindi
  - Regularized stage: <NOM, ALL> Akhvakh, Karata
- pressure to conform to the constraint against nominativeless frames

5 *bite* across Andic languages

- *bite* is used in 3 case frames across Andic languages
  - <ERG, NOM, LOC> in Tindi *saldi bix̄^{j}iɬ^{j}a*, Y_{NOM} = *saldi* `teeth'

(15) χ^{w}āj-i di-č'-i sadi b-ix̄^{j}-o. Tindi
dog-ERG 1SG$_0$-TPL-LOC tooth.PL N-hold-PF
The dog bit me. *(Sh. Magomedov, p.c.)*

  - <ERG, LOC> in Akhvakh *q̄'eleč'uruɪa*, Chamalal *q̄'āna*, Karata *q̄^{w}arała*

(16) χ^{w}aj-ol q̄'^{w}ar-e di-č'o. Karata
dog-ERG bite-PF 1SG$_0$-TPL[LOC]
The dog bit me.

(17) χ^{w}e-de di-g-e q̄' eleč'-ari. Akhvakh
dog-ERG 1SG$_0$-TPL-LOC bite-PF
The dog bit me.

(18) χ^{w}aj-d q̄'ān-nida o-sūu-č'. Chamalal
dog-ERG bite-PF DEM-M$_0$-TPL[LOC]
The dog bit him.

  - <ERG, NOM> in Bagvalal *salīla*, Godoberi *q̄'ami*

(19) ʕamo-r salī di-b lela. Bagvalal
donkey$_0$-ERG bite.PF$_1$ 1SG$_0$[GEN]-N hand
The donkey bit my hand.

(20) χ^{w}aji-di q̄'am-i ho-w. Godoberi
dog-ERG bite-PF$_1$ DEM-M
The dog bit him.

- Bagvalal *salīla* `bite' <ERG, NOM> is cognate with `tooth' in:
  - Bagvalal & Chamalal *sal*^{w}
  - Karata *sale*
  - Tindi *salu*
  - Andi *sol*

(21) Plausible etymology of `bite' in Bagvalal
sal^{w} + V $\succ$ *salīla*

- Godoberi *q̄'ami* <ERG, NOM>, Chamalal *q̄'āna* <ERG, LOC> can be reconstructed to **q̄'am*
- case frame difference is not due to two different etymologies, change must have occurred in the case frame of at least one of them
- reconstruction of the evolution of the case frame associated to *bite*
  - Initial stage: <ERG, NOM, LOC> Tindi
  - Nomativeless stage: <ERG, LOC> Akhvakh, Chamalal, Karata
  - Regularized stage: <ERG, NOM> Bagvalal, Godoberi
- pressure to conform to the constraint against nominativeless frames

6 Semantically motivated change of case

6.1 Faithfulness to inherent cases

- Two ways to satisfy nominative constraint: is the choice random?
- No: which case-marked argument remains *un*changed depends on its semantic role
- ERG and LOC/ALL are cases associated with specific semantic roles (i.e. they are `inherent cases' Woolford (2006))
  - ERG is associated with the Agent semantic role
  - LOC/ALL is associated with the Target (or Goal) semantic role

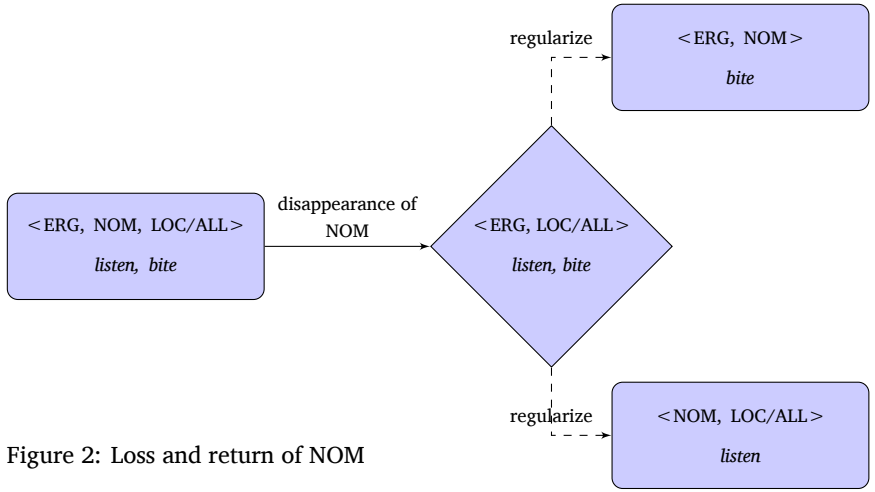

Figure 2: Loss and return of NOM

- at the nominativeless stage, *listen* is expressed in $<X_{ERG}, Y_{LOC/ALL}>$
  - in the case frame associated to *listen*, Y is the target of X's attention
  - $Y_{target} \rightarrow Y_{LOC/ALL}$ (inherent case)
  - the case on $Y_{LOC/ALL}$ is `protected', and the one on X_{ERG} is more likely to change
- at the nominativeless stage, *bite* is expressed in $<X_{ERG}, Y_{LOC/ALL}>$
  - in the case frame associated to *bite*, X is the agent
  - $X_{agent} \rightarrow X_{ERG}$ (inherent case)
  - the case on X_{ERG} is `protected', and the one on $Y_{LOC/ALL}$ is more likely to change
- because `real agents' are linked to the semantics of ERG, and because targets are linked to the semantics of LOC/ALL, they are `protected' from the effect of the nominative constraint
- the constraint against nominativeless case frames is `blindly' active
- we see the effect of the nominative constraint on arguments whose semantic role is not inherently linked to the case they happen to bear

6.2 On the semantic import of NOM

- ERG and LOC/ALL, as inherent cases, are clearly linked to specific semantic values
- in $<X_{ERG}, Y>$, NOM is linked to the notion of affectedness
  - learning argument structure depends on affectedness (Gropen et al. 1991)
  - role of affectedness in the dative alternation in English (Krifka 2013)

(22) Affectedness
X affects Y iff X causes Y to change.

- affectedness is gradable
  $Y_1 >_{affected} Y_2$ iff Y_1 has undergone more change (from its initial state) than Y_2 has

(23) In the case frame $<X_{ERG}, Y>$,
where Y = NOM
or
Y = LOC/ALL
the referent of $Y_{NOM} >_{affected}$ the referent of $Y_{LOC/ALL}$.

- evidence from reflexes of **q̄'am*

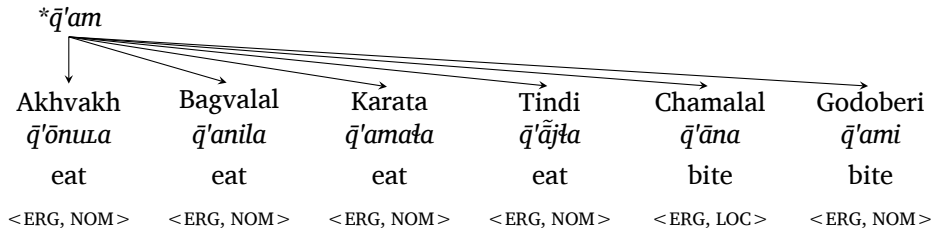

Figure 3: Reflexes of **q'am*

(24) Generalization:
reflex of **q̄'am* = `eat' → <ERG, NOM>

- evidence from case alternations
  - Akhvakh *q̄'eleč'uruɬa* `bite'
    - * <ERG, LOC>

(25) wašo-de ʕeče-g-e q̄'eleč'-ari.
boy$_0$-ERG apple-TPL-LOC bite-PF
The boy bit into the apple (clung to it). *(I. Abdulaeva, p.c.)*

    - * <ERG, NOM>

(26) wašo-de ʕeče q̄'eleč'-ari.
boy$_0$-ERG apple bite-PF
The boy bit off a chunk of apple.

  - Karata *c̄'imała* `pinch'
    - * <ERG, LOC>

(27) ho-šu-l di-č'o c̄'im-e
DEM-M$_0$-ERG 1SG$_0$-TPL[LOC] pinch-PF
He pinched me.

    - * <ERG, NOM>

(28) ho-šu-l bušīdi c̄'im-e
DEM-M$_0$-ERG berries pinch-PF
He crushed berries (into jam) *(R. Khalidova, p.c.)*

- alternations in the case frame of *bite* and *pinch* show us 2 things:
  - case change to NOM is first driven by really affected Y arguments while less affected Y arguments are last to shift (this may be the reason why the verbs meaning `eat' appear with <ERG, NOM>, but there is still some variation with those meaning `bite')

– in the <ERG, NOM> case frame, ERG is not the only case with semantic import

(29) Conclusion:
To the extent that we need affectedness to explain these two patterns, it is plausible that it is also at play in the regularization of case frames.

7 Summary

- A1: the coalescence of nominative argument and V yields (accidental) `exceptional' valency frames
- A2: `exceptional' valency frames undergo regularization by changing one case to NOM
- A3: which argument/case, X_{ERG} or $Y_{LOC/ALL}$, remains *un*changed depends on the semantic role of X, Y & with V that keep X_{ERG}, Y's shift to NOM happens first in uses of V in which Y is most affected (*e.g.* `pinch')

Abbreviations

H^+ = human plural, N^+ = non-human plural, $_0$ = oblique stem, PL = plural, SG = singular, NOM = nominative, ERG = ergative, TPL = topological formative, LOC = locative, ALL = allative, ABL = ablative, COP = copula, IPF = imperfective, PF = perfective, N = neuter, CAUS = causative, ADV = adverb, DEM = demonstrative, GEN = genitive, INF = infinitive, NEG = negation

References

Creissels, D. (2009). Valency properties of Northern Akhvakh verbs. http://www.deniscreissels.fr/public/Creissels-valency_classes_project_Akhvakh.pdf.

Gropen, J., S. Pinker, M. Hollander, and R. Goldberg (1991). Affectedness and direct objects: the role of lexical semantics in the acquisition of verb argument structure. *Cognition*.

Krifka, M. (2013). Semantic and pragmatic conditions for the dative alternation. http://amor.cms.hu-berlin.de/~h2816i3x/Publications/DativeAlternationKorea.pdf.

Magomedova, P. (1999). *Chamalinsko-ruskij slovar' (Chamalal-Russian dictionary)*. Makhachkala: Dagestanskij Nauchnyj Centr Rossiskoj Akademii Nauk.

Magomedova, P. (2003). *Tindinsko-ruskij slovar' (Tindi-Russian dictionary)*. Makhachkala: Dagestanskij Nauchnyj Centr Rossiskoj Akademii Nauk.

Magomedova, P. (2004). *Bagvalinsko-ruskij slovar' (Bagvalal-Russian dictionary)*. Makhachkala: Dagestanskij Nauchnyj Centr Rossiskoj Akademii Nauk.

Magomedova, P. and I. Abdulaeva (2007). *Akhvakhsko-ruskij slovar' (Akhvakh-Russian dicitonary)*. Makhachkala: Dagestanskij Nauchnyj Centr Rossiskoj Akademii Nauk.

Magomedova, P. T. and R. S. Khalidova (2001). *Karatinsko-ruskij slovar'. (Karata-Russian dictionary.)*. Makhachkala: Dagestanskij Nauchnyj Centr Rossiskoj Akademii Nauk.

Pasquereau, J. (2010). Spatial cases in karata: spatial and non-spatial uses. Master's thesis, Université Lyon II.

Pasquereau, J. (2011). Valency in Karata: a preliminary study. Master's thesis, Université Lyon II.

Saidova, P. (2006). *Godoberinsko-ruskij slovar' (Godoberi-Russian dictionary)*. Dagestanskij Nauchnyj Centr Rossiskoj Akademii Nauk.

Woolford, E. (2006). Lexical case, inherent case, and argument structure. *Linguistic Inquiry 37*, 111--130.